AC POWER METER

BACK GROUND PROJECT , UG4 , ELECTRICAL ENGINEERING

TEAM MEMBERS

DWAIPAYAN SHIKARI , ROLL NO - 002210801231

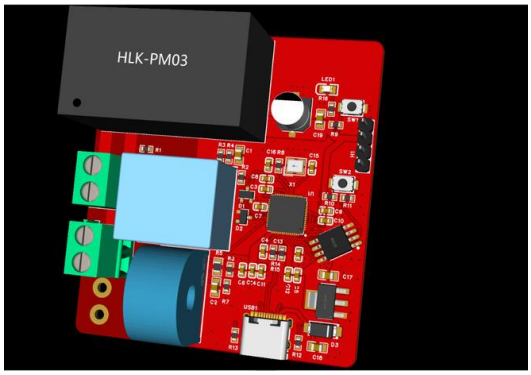
SHRAMANA MONDAL , ROLL NO - 002210801239

PRATTAY BARUA, ROLL NO - 002210801246

JAYRUP DAS , ROLL NO - 002210801131

WHAT DONE TILL NOW..

FIRST VERSION --> TESTING --> GOT BORED --> PCB DESIGN --> THIS BUILD --> STILL INCOMPLETE BUT WORKS

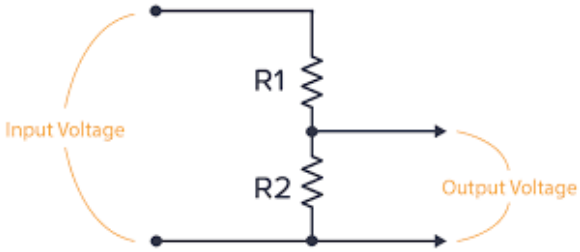


THE FIRST VERSION

IT WAS BASIC, MAIN OPERATING PRINCIPLE WAS TO MEASURE FROM ZMPT VOLTAGE SENSOR AND CURRENT TRANSFORMER

SCHEME WAS

ZMPT MODULE CONVERTS AC 230V TO 0-5V LEVEL , 2.5V WAS ZERO , BUT IN OUR CASE WE USED A SIMPLE VOLTAGE DIVIDER, 2/3 RATIO , TO FEED IT INTO ESP32 3.3V



THEN , SIMPLE CURRENT TRANSFORMER , THE LOAD WIRE LOOP IS PASSED , AND CURRENT INDUCED , THEN A BLEED RESISTOR TO CONVERTS IT INTO VOLTAGE

QUICK , SIMPLE , ROBUST , THEN FOR ACCESS WE USED ESP32'S WIFI PORT , WE MADE A HTTP BASED WEB SERVER , TO SEE LIVE VOLTAGE AND CURRENT DATA

TESTING PHASE

actual(v)	calculated (v)	current (amp)	current_calculated (amp)
220	211	0.32	0.46
210	204	0.31	0.44
200	192	0.3	0.4
190	182	0.29	0.4
180	173	0.28	0.39
170	163	0.27	0.37
160	154	0.26	0.35
150	143	0.25	0.32
140	133	0.24	0.3
130	124	0.23	0.29
120	115	0.22	0.222
110	105	0.21	0.21
100	96	0.19	0.185
90	85	0.18	0.178
80	77	0.17	0.157
70	67	0.16	0.13
60	57	0.14	0.125
50	47	0.13	0.12
40	37	0.11	0.08
30	28	0.09	0.06
20	18	0.07	0.047
10	9	0.05	0.02
0	0	0.01	0

GOT SOME ERROR , VOLTAGE WAS ALMOST OK , CURRENT READING WAS WEAK THOUGH , NAIVELY CHOOSE HIGHEST TURNS RATIO CURRENT TRANSFORMER ,SO , SIGNAL TO NOISE RATIO WAS HIGH

ANY WAY THE REVISED FIRMWARE HAS THESE THINGS CALIBRATED

GOT BORED

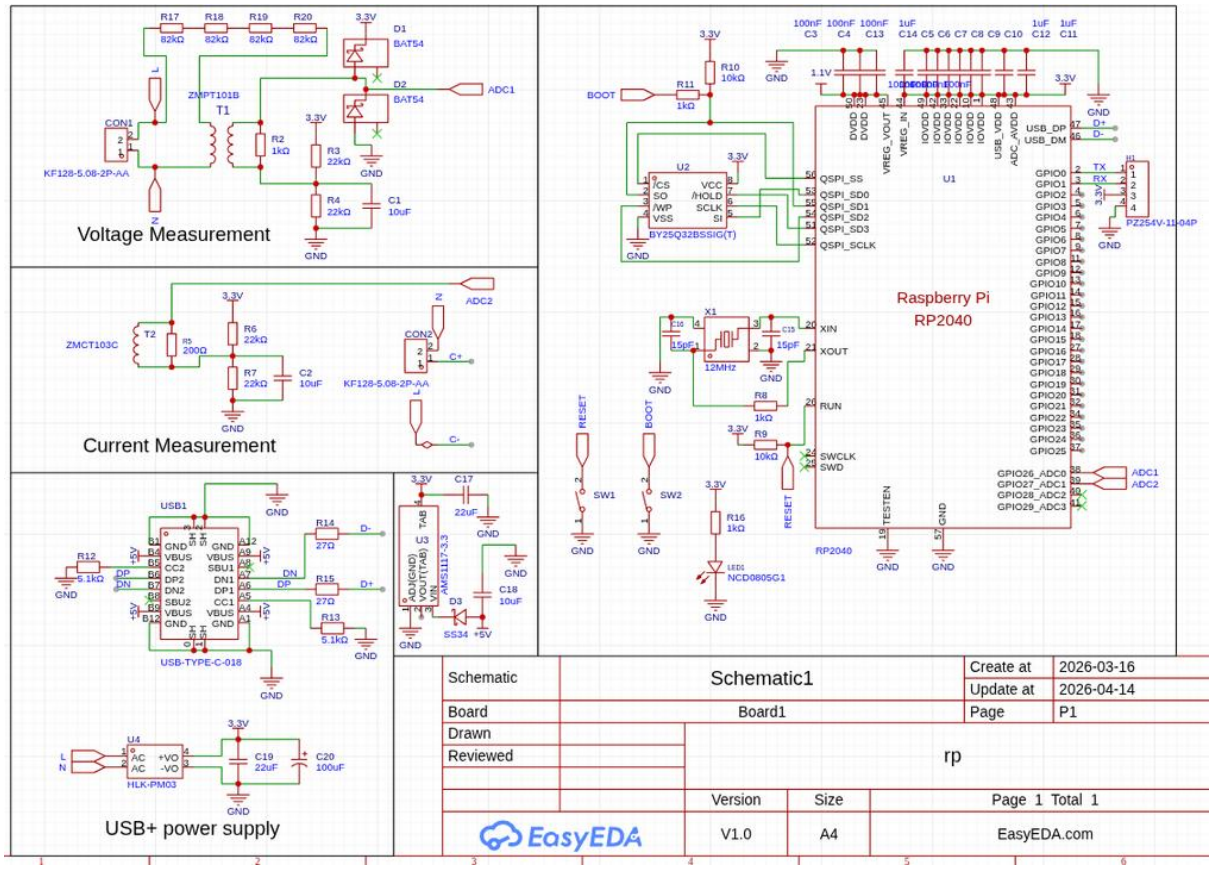
WITH THINGS DONE WITHIN FEW WEEKS , TEAM MEMBERS GREW BORED

BUT BOREDOM BRINGS PURPOSE

SO STARTED TO DESIGN A PCB TO FIT ALL COMPONENTS

BUT THE ADC OF ESP32 IS NOISY SO , CHOOSE A SIMPLER HIGH PERFORMANCE MCU , RP2040 , 12 BIT (BUT USING MICROPYTHON SO SCALE IS SHIFTED TO 0-65535)

HERE IS THE SCHEMATIC...



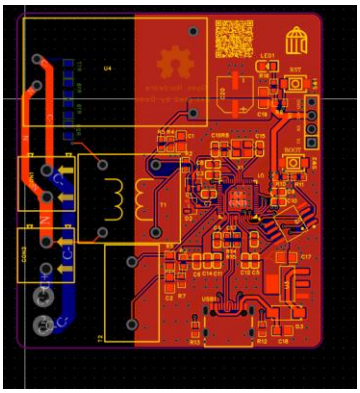
VOLTAGE MEASUREMENT(DC 1.65 V BIAS+AC SWING+PIN PROTECTION)

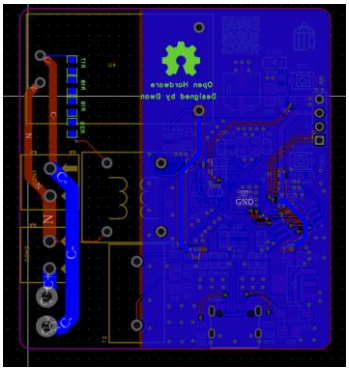
CURRENT MEASUREMENT(DC 1.65V BIAS+AC SWING)

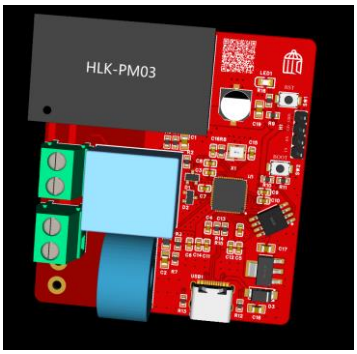
USB+ AC TO DC POWER SUPPLY

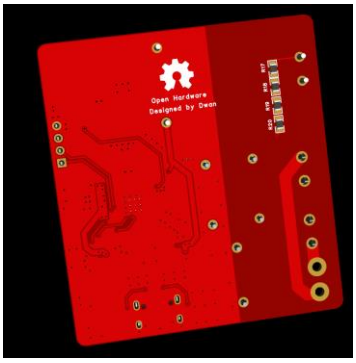
PCB DESIGN

DONE ON EASYEDA PRO SOFTWARE



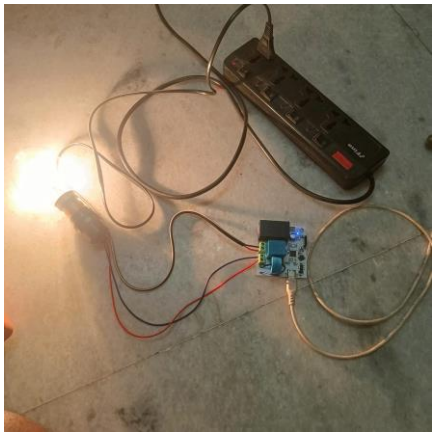






MANUFACTURED USING JLCPCB CHINA(PCB ONLY), THEN HAND SOLDERED MANUALLY

THIS BUILD



power_meter.py

```python
import utime
import math
from machine import ADC, Pin

## Configuration ---------------------------------------------------------------
SAMPLES = 500
SAMPLE_DELAY_US = 200
ADC_VREF = 3.3
ADC_COUNTS = 65335

## Calibration -----------------------------------------------------------------
VCAL = 268.97
ICAL = 16.667 / 3

## Hardware Init ---------------------------------------------------------------
adc_v = ADC(Pin(26))
adc_i = ADC(Pin(27))

def print_header():
```

```
MPY: soft reboot
================================================================================
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                POWER METER V.0
        TEAM MEMBERS:
        1)Dwaipayan Shikari
        2)Shramana Mondal
        3)Prattay Barua
        4)Jayrup Das
```

VERY SIMPLE PROCESS YET AGAIN

CHOOSE A SAMPLING RATE OF 5KHZ WITH 500 SAMPLES PER SECOND, EACH 20MS WINDOW OR HALF GET 10 SAMPLES , MORE THAN ENOUGH TO GET PROPER , ALTHOUGH IF WE AIM FOR HARMONICS AND OTHER STUFF , JUST HAVE INCREASE SAMPLING FREQUENCY AND SAMPLES

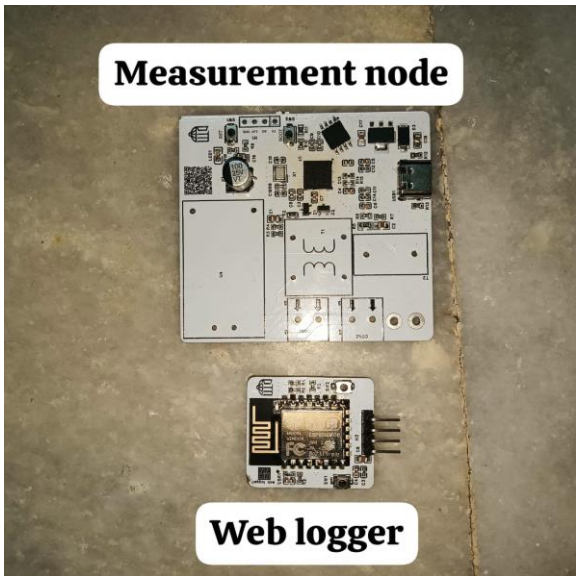
RAW READINGS -->ADDED --> GET OFFSET --> SUBTRACT FROM ORIGINAL --> GET TRUE AC READING --> SQUARED --> RMS

ISSUES STILL TO FIX AND BEHIND THE SCENES

THE SOLDERING OF RP2040 IS CAUSING FEW ISSUES, SINCE IT IS MANUALLY SOLDERED WITHOUT ANY SOLDER PASTE OR STENCIL , JUST USING LEADED SOLDER AND HOT AIR , FEW CONNECTIONS ARE SUPPOSEDLY OF THE PLACE , SO ON THE VOLTAGE PIN THE

READING IS WRONG , BUT CURRENT PIN IS PERFECT

ALSO INITIALLY PLANNED TO PUT A WEB LOGGER TOO, BUT OF DESIGN NAIVELY DID NOT PUT ANY TESTPADS AND SOLDERING COULD STILL BE AN ISSUE

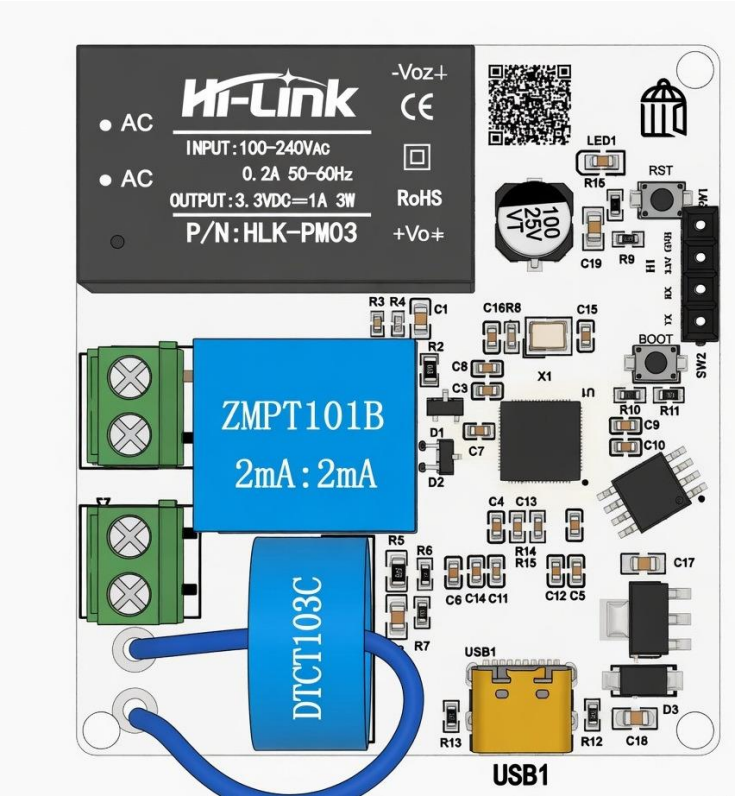


INITIAL PLAN

ALSO LOGISTICS WAS TROUBLESOME TOO , SINCE FROM PREVIOUS EXPERIENCES HAD TO DEAL WITH CUSTOMS , SO THIS TIME CHOOSE AN INDIAN BROKER , COSTS LESS BUT LONG TIME , ALSO

FEW INDIAN MANUFACTURES ARE THERE BUT NOT AS GOOD QUALITY AS JLCPCB

MODELS AND FIRMWARES



ALL THE TECHNICAL DETAILS ARE IN

GITHUB.COM/DWAN6767/UNIOPME

COSTS AND TIME TAKEN TO BUILD

1 WEEK OF DESIGN + 2-3 WEEKS OF GETTING + 2-3 DAYS :1 MONTH

SOLDERING AND TESTING

TOTAL COST OF PROJECT (EXCLUDING SOLDERING , ELECTRICITY AND FREE LABOUR) : 2500 -3000

CONTRIBUTIONS (BOTH VERSIONS)

DWAIPAYAN SHIKARI (PCB DESIGN + PART OF LOGISTICS AND SOLDERING)

SHRAMANA MONDAL (FIRMWARE)

PRATTAY BARUA (LOGISTICS , COMPONENT SOURCING ,TESTING)

JAYRUP DAS (SOLDERING ,TESTING , 3D MODELS)